Final Exam Review

Cumulative: Lessons 1–18, Emphasis on Lessons 10–19

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Today's Agenda

- Exam logistics and structure
- First-half foundations (what's on the exam from Lessons 1–9)
- Second-half concepts — the main focus:
  - Data exploration
  - Paired t-test
  - ANOVA and post-hoc testing
  - Categorical data analysis
  - Nonparametric tests
  - Correlation and simple linear regression
  - Power, sample size, and synthesis
- Choosing the right test
- Your questions

The final exam opens TODAY at 3:00 PM and closes Friday, March 20 at 5:00 PM

Exam Logistics

Exam Details

Format:

- 8 parts, 150 points total
- Designed for 4–6 hours completion
- Mix of conceptual questions, R code, and interpretation

Resources allowed:

- ✅ Course materials (slides, homework, textbook)
- ✅ R documentation and help files (`?t.test`, `?pnorm`, etc.)
- ✅ Your own notes
- ❌ Other students
- ❌ AI assistants (ChatGPT, Claude, etc.)
- ❌ Online forums or external help

Submit both `.qmd` and `.html` files — naming: `Lastname_FirstInitial_Final`

What's Covered

Part 1: Foundations — first half (15 pts)

Part 2: Exploring the data (15 pts)

Part 3: Paired t-test (20 pts)

Part 4: ANOVA and post-hoc (20 pts)

Part 5: Categorical data (20 pts)

Part 6: Nonparametric tests (15 pts)

Part 7: Correlation and SLR (25 pts)

Part 8: Power, sample size, synthesis (20 pts)

Key dataset: `diabetes_trial.csv`

- Simulated RCT, 180 patients, 3 treatment arms (Metformin, Lifestyle, Combined), 6-month follow-up
- Read the variable descriptions carefully before starting

General Tips

Before you start:

- Set up your workspace: load packages, read in data
- Make sure code chunks evaluate
- Check that `here("data", "your_file.csv")` finds your file

While working:

- Interpret everything in complete sentences — reference the numbers
- Use all 6 steps for hypothesis tests
- Use comments in your R code to show your thinking
- **Save often**

Before submitting:

- Render and confirm no errors
- Verify both `.qmd` and `.html` are submitted
- Check file naming format

Part 1: First-Half Foundations

What from the first half is on the exam?

Part 1 (15 points) tests three targeted areas from Lessons 1–9:

1.1 — Study design (5 pts)

- Experiment vs. observational study
- Why randomization matters
- When can we claim causation?

1.2 — Bayes' theorem / PPV (5 pts)

- Given sensitivity, specificity, prevalence → calculate PPV
- Interpret PPV in context

1.3 — Normal distributions and CLT (5 pts)

- `pnorm()` for an individual value
- `pnorm()` for a sample mean (using $SE = \frac{\sigma}{\sqrt{n}}$)

Tip

Your **midterm review slides** cover all of this in detail — a quick re-read is the best preparation for Part 1.

Study design: the essentials

Experiment:

- Researchers **assign** treatments (random assignment)
- Controls for confounding
- Can establish **causation**

Observational study:

- Researchers observe without interfering
- Cannot control for all confounders
- Can only show **association**

Causation checklist

A study can claim causation **only if:**

1. It is a randomized experiment
2. Random assignment was used
3. Other explanations have been ruled out

Key question: Was treatment *assigned* by the researchers, or did subjects choose/arrive with it?

Bayes' Theorem: PPV setup

Given sensitivity, specificity, and prevalence:

$$PPV = P(\text{Disease} \mid \text{Test}+) = \frac{\text{sensitivity} \times \text{prevalence}}{\text{sensitivity} \times \text{prevalence} + (1 - \text{specificity}) \times (1 - \text{prevalence})}$$

Worked example: A TB screening test has sensitivity = 0.79, specificity = 0.87, applied in a population with 6% prevalence.

```r
sensitivity <- 0.79
specificity <- 0.87
prevalence  <- 0.06

ppv <- (sensitivity * prevalence) /
  (sensitivity * prevalence + (1 - specificity) * (1 - prevalence))
ppv
```

```
[1] 0.2794811
```

Tip

Interpreting PPV: "Of all patients who test positive, ___% actually have the condition."

A high PPV → test is useful for **confirming** disease. A low PPV → test is better for **screening** (ruling out).

Normal distributions and CLT: two scenarios

Individual observation: $X \sim N(\mu, \sigma^2)$

```r
# LDL cholesterol: mu = 130, sd = 35
# P(LDL > 160) — "borderline high"

pnorm(160, mean = 130, sd = 35,
      lower.tail = FALSE)
```

```
[1] 0.195683
```

The SD here is σ — variability of **individuals**

Sample mean: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

```r
# for n = 25, same population
# P(Xbar > 140)

se <- 35 / sqrt(25)
pnorm(140, mean = 130, sd = se,
      lower.tail = FALSE)
```

```
[1] 0.07656373
```

The variability here is measure by $SE = \sigma/\sqrt{n}$ — variability of **sample means**

The key distinction

When the question says "a random individual" → use σ.

When the question says "a sample of n" → use $SE = \sigma/\sqrt{n}$.

Part 2: Exploring the Data

Descriptive statistics by group

```r
# Overall summary of a continuous variable
nhanes.samp %>%
  summarise(
    n      = sum(!is.na(Pulse)),
    mean   = mean(Pulse, na.rm = TRUE),
    sd     = sd(Pulse, na.rm = TRUE),
    median = median(Pulse, na.rm = TRUE)
  )

# By group
nhanes.samp %>%
  group_by(Gender) %>%
  summarise(
    n    = sum(!is.na(Pulse)),
    mean = mean(Pulse, na.rm = TRUE),
    sd   = sd(Pulse, na.rm = TRUE)
  )

# Or with rstatix
nhanes.samp %>%
  group_by(Gender) %>%
  get_summary_stats(Pulse, type = "mean_sd")
```

Visualization: histograms and boxplots

```r
# Histogram — check shape, skewness, outliers
ggplot(nhanes.samp,
       aes(x = BPSys1)) +
  geom_histogram(bins = 30,
                 color = "white") +
  labs(x = "Systolic BP (mmHg)", y = "Count",
       title = "Distribution of systolic blood pressure")

# Boxplots by group with individual points
ggplot(nhanes.samp,
       aes(x = Gender, y = Pulse, fill = Gender)) +
  geom_boxplot(alpha = 0.6,
               width = 0.5) +
  geom_jitter(width = 0.2, alpha = 0.4) +
  labs(x = "Gender", y = "Pulse (bpm)",
       title = "Pulse rate by gender")
```

Tip

`geom_jitter()` is layered **after** `geom_boxplot()` so points appear on top. Always include `width = 0.2` or similar to prevent overplotting.

Frequency tables with `tabyl()`

```r
# One-way table
nhanes.samp %>%
  tabyl(SmokeNow)

# Two-way table with row percentages
nhanes.samp %>%
  tabyl(PhysActive, SmokeNow) %>%
  adorn_totals(where = "row") %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

Tip

`adorn_percentages("row")` gives **row percentages** — the proportion *within* each row group. `adorn_ns()` keeps the raw counts alongside so both n and % are visible.

Part 3: Paired t-test

When is a paired design appropriate?

Paired (dependent) design:

- Same subject measured **twice** (before/after)
- Matched pairs (sibling controls, eye studies)
- The pairing controls for between-subject variability

Recognize it by:

- the same ID appears in both measurements
- there is a natural link between observations

Why pairing matters

- If you ignore the pairing and treat the two measurements as independent groups, you **throw away information** about within-subject change.
- The paired test uses differences $d_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$ and is equivalent to a one-sample t-test on those differences.
- Ignoring pairing → larger SE → less power → more likely to miss a real effect.

Paired t-test: the workflow

Step 1: Create the difference variable

```r
# Example: SBP before and after a 12-week exercise intervention
bp_data <- bp_data %>%
  mutate(sbp_change = sbp_pre - sbp_post)  # positive = decrease in BP
```

Step 2: Run the test

```r
# Option A: one-sample t-test on differences
t.test(bp_data$sbp_change, mu = 0, alternative = "two.sided")

# Option B: paired t-test directly
t.test(bp_data$sbp_pre, bp_data$sbp_post,
       paired = TRUE, alternative = "two.sided")
```

Tip

Both give identical results. Option A is often clearer for applying the six steps — you are explicitly testing whether the mean difference equals zero.

Paired t-test: the six steps

Step 1: $H_0 : \mu_d = 0$ (no mean change) vs. $H_A : \mu_d > 0$ (BP decreased, if directional)

Step 2: $\alpha = 0.05$

Step 3: Assumptions

- Independence: one pair per subject ✓ (study design)
- Normality of differences: large n → CLT; small n → check histogram or QQ plot of differences

Step 4: $t = \dfrac{\bar{d} - 0}{s_d/\sqrt{n}}$ — reported as `t` in `t.test()` output

Step 5: p-value from `t.test()` output; $df = n - 1$

Step 6: Reject if p < 0.05. Report in context:

> "There was a statistically significant decrease in SBP from baseline to 12 weeks (t = ___, df = ___, p = ___). On average, SBP decreased by ___ mmHg (95% CI: ___ to ___)."

Part 4: ANOVA and Post-Hoc Testing

When to use one-way ANOVA

ANOVA: comparing means across 3+ groups

Use one-way ANOVA when:

- **Outcome:** continuous
- **Predictor:** categorical with **3 or more groups**
- **Question:** Do any of the group means differ?

Why not just run multiple t-tests?

- Each t-test has a 5% Type I error rate.
- With 3 pairwise comparisons, the family-wise error rate inflates to ~14% — you're much more likely to find a spurious significant result by chance alone.
- **ANOVA tests all groups simultaneously**, preserving the overall error rate at α.

ANOVA assumptions (1/3)

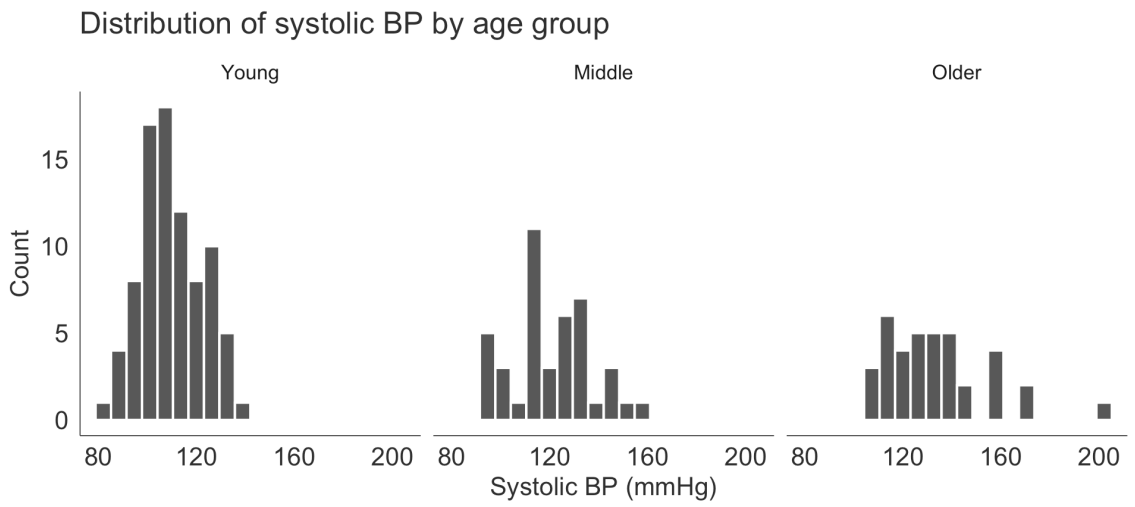
1. **Independence:** one observation per subject (study design) ✓
2. **Normality:** approximately normal within each group
3. **Equal variance:** SD ratio (max SD / min SD) should be < 2

```r
# Create 3 age groups, then check group-level stats
nhanes_age <- nhanes.samp %>%
  mutate(age_group = cut(Age,
                         breaks = c(0, 35, 55, 100),
                         labels = c("Young", "Middle", "Older"))) %>%
  filter(!is.na(age_group), !is.na(BPSys1))
```

ANOVA assumptions (2/3): normality

Check: distributions approximately normal within each group

```
ggplot(nhanes_age, aes(x = BPSys1)) +
  geom_histogram(bins = 20, color = "white") +
  facet_wrap(~ age_group) +
  labs(x = "Systolic BP (mmHg)", y = "Count",
       title = "Distribution of systolic BP by age group")
```



Tip

With reasonably large groups, moderate departures from normality are acceptable — the CLT helps. Look for severe skew or extreme outliers.

ANOVA assumptions (3/3): equal variance

Check: SD ratio (max SD / min SD) should be < 2

```r
nhanes_age %>%
  group_by(age_group) %>%
  summarise(n    = n(),
            mean = mean(BPSys1),
            sd   = sd(BPSys1)) %>%
  mutate(sd_ratio = max(sd) / min(sd))
```

```
# A tibble: 3 × 5
  age_group     n  mean    sd sd_ratio
  <fct>     <int> <dbl> <dbl>    <dbl>
1 Young        84  110.  12.4     1.68
2 Middle       42  120.  16.2     1.68
3 Older        37  133.  20.8     1.68
```

Tip

SD ratio < 2 → equal variance assumption is met ✓

If the ratio is borderline (say, 1.8–2.5), note the concern but proceed — ANOVA is fairly robust to mild violations when group sizes are similar.

ANOVA: the six steps + R code

Step 1: $H_0 : \mu_1 = \mu_2 = \mu_3$ vs. H_A : at least one group mean differs

Steps 2–3: $\alpha = 0.05$; check assumptions ✓

Steps 4–5: Fit and summarize

```r
anova_fit <- aov(BPSys1 ~ age_group, data = nhanes_age)

# Traditional output
summary(anova_fit)

# Tidy output
broom::tidy(anova_fit)
```

Step 6: Report F-statistic with **both** df values and p-value

> "There was a significant difference in mean systolic blood pressure between at least two age groups (F(2, 160) = ____, p = ____)."

Post-hoc testing: Tukey's HSD

If ANOVA rejects H_0, use Tukey HSD to identify **which pairs** differ:

```
rstatix::tukey_hsd(anova_fit)
```

Reading the output:

- `estimate` = difference in group means
- `p.adj` = adjusted p-value (controls family-wise error rate)
- Report pairs where `p.adj < 0.05`

Writing a results paragraph

Include: descriptive stats for each group + F-statistic + p-value + specific post-hoc comparisons.

> "Mean [outcome] was ___ (SD = ___) in Group A, ___ (SD = ___) in Group B, and ___ (SD = ___) in Group C. A one-way ANOVA indicated a significant difference between group means (F(df1, df2) = ___, p = ___). Tukey post-hoc tests revealed that [Group A vs. Group C: mean diff = ___, p = ___] differed significantly, while [other comparisons] did not."

Part 5: Categorical Data Analysis

Chi-squared test for independence

When to use: Both variables are **categorical**, testing whether they are associated

```r
# Step 1: Build the contingency table
tab <- nhanes.samp %>%
  filter(!is.na(SmokeNow), !is.na(PhysActive)) %>%
  tabyl(PhysActive, SmokeNow)

# Step 2: Check expected counts — all should be ≥ 5
chisq.test(tab)$expected

# Step 3: Run the test
chisq.test(tab)
```

Hypotheses (in words):

- H_0: There is no association between physical activity and smoking status
- H_A: There is an association between physical activity and smoking status

Two-proportion z-test

When to use: Exactly two groups, binary outcome

```r
# Compute successes and group sizes
counts <- nhanes.samp %>%
  filter(!is.na(SmokeNow)) %>%
  group_by(Gender) %>%
  summarise(
    success = sum(SmokeNow == "Yes", na.rm = TRUE),
    n       = sum(!is.na(SmokeNow))
  )

# Two-proportion test
prop.test(x = counts$success,
          n = counts$n,
          alternative = "two.sided")
```

Hypotheses:

- $H_0: p_1 = p_2$
- $H_A: p_1 \neq p_2$

Chi-squared vs. two-proportion test

	Chi-squared	Two-proportion (`prop.test`)
Groups	2 or more	Exactly 2
Outcome	Categorical (any # levels)	Binary (success/failure)
Hypotheses	Independence vs. association	$p_1 = p_2$ vs. $p_1 \neq p_2$
Output	χ^2, df, p-value	z (or χ^2), p-value, **CI for difference**

Tip

For exactly **2 groups and a binary outcome**, both tests give equivalent p-values. Prefer `prop.test()` when you want a **confidence interval for the difference in proportions** — that CI gives more information than the p-value alone.

Part 6: Nonparametric Tests

When to go nonparametric

The decision is about the data, not just sample size

Use a nonparametric test when:

- The distribution is **heavily skewed** or has extreme outliers
- The outcome is **ordinal** (pain scales, severity scores)
- Sample sizes are small **and** normality is implausible

How to assess: plot the data first (histogram or boxplot); check the SD ratio across groups if comparing 3+ groups

Warning

Small n alone does not automatically mean nonparametric. If the data look approximately normal, the parametric test is valid and more powerful. The criterion is whether the **assumptions** are plausibly met.

Three nonparametric tests: the map

Parametric test	Nonparametric alternative	R function
One-sample / paired t-test	Wilcoxon signed-rank	`wilcox.test(x, mu = 0)`
Two-sample t-test	Wilcoxon rank-sum (Mann-Whitney)	`wilcox.test(x, y)`
One-way ANOVA	Kruskal-Wallis	`kruskal.test()`

Hypotheses (state correctly!):

- H_0: The distributions of [outcome] are **identical** across groups
- H_A: At least one group has a different distribution

Important

Do **not** state hypotheses in terms of medians being equal — this is only valid under additional assumptions. "Distributions are identical" is the correct framing.

Kruskal-Wallis + pairwise follow-up

```r
# Kruskal-Wallis test
kw_result <- kruskal.test(Pulse ~ age_group, data = nhanes_age)

kw_result %>%
  broom::tidy()

# Pairwise Wilcoxon with Bonferroni correction (if K-W is significant)
pairwise.wilcox.test(nhanes_age$Pulse, nhanes_age$age_group,
                     p.adjust.method = "bonferroni") %>%
  broom::tidy()
```

Tip

There is no nonparametric equivalent of Tukey HSD. Bonferroni correction is a straightforward choice — it controls the family-wise error rate across all pairwise comparisons. With only 3 pairs here, the penalty for using Bonferroni is small.

Part 7: Correlation and Simple Linear Regression

This section is 25 points — the largest on the exam

No homework covered this material

Part 7 is the only section that was not practiced in a homework assignment. The exam provides extra guidance (hints in code chunks). Read the questions carefully and follow the steps.

The material maps directly to the Lesson 21 slides — a review of those slides before the exam is time well spent.

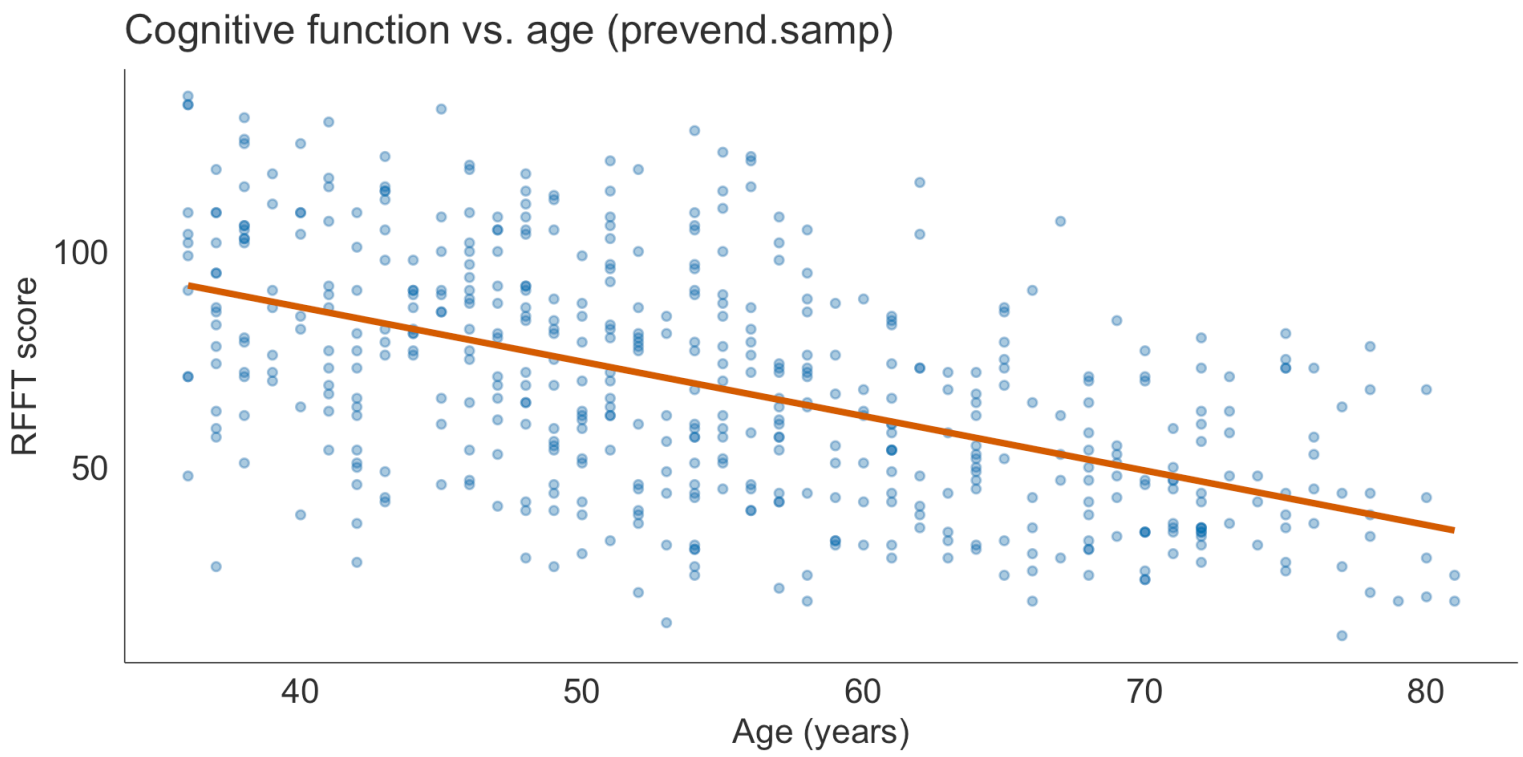
What Part 7 tests:

- Scatterplot + regression line
- Pearson and Spearman correlation: `cor()`
- Fitting `lm()`, interpreting `tidy()` output
- LINE assumption checks with `augment()` + diagnostic plots
- R^2 from `glance()`
- Prediction vs. confidence interval: `predict()`

Scatterplot and correlation

Using `prevend.samp`: does cognitive function (`RFFT` score) decline with age?

```
# Scatterplot with regression line
ggplot(prevend.samp, aes(x = Age, y = RFFT)) +
  geom_point(alpha = 0.4, color = "#0072B2", size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, color = "#D55E00") +
  labs(x = "Age (years)", y = "RFFT score",
       title = "Cognitive function vs. age (prevend.samp)")
```



Correlation coefficients

```r
# Pearson: strength of LINEAR association
cor(prevend.samp$Age, prevend.samp$RFFT,
    use = "complete.obs")
```

```
[1] -0.5338617
```

```r
# Spearman: rank-based, robust to skew/outliers
cor(prevend.samp$Age, prevend.samp$RFFT,
    method = "spearman",
    use = "complete.obs")
```

```
[1] -0.5318226
```

Describing the relationship: direction (positive/negative), form (linear/curved), strength (weak/moderate/strong), unusual observations

Tip

The Pearson and Spearman values are usually close when the relationship is approximately linear without extreme outliers. A large gap suggests outliers or nonlinearity are influencing the Pearson r.

Fitting the regression model

```r
# Fit
rfft_model <- lm(RFFT ~ Age, data = prevend.samp)

# Coefficients, SE, t, p, CI
tidy(rfft_model, conf.int = TRUE)
```

```
# A tibble: 2 × 7
  term        estimate std.error statistic   p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
1 (Intercept)   138.      5.02        27.4 1.41e-101   128.      147.
2 Age            -1.26    0.0895     -14.1 3.50e- 38    -1.44     -1.09
```

Interpreting the slope ($\hat{\beta}_1$):

> "For every 1-year increase in age, RFFT score is expected to change, on average, by $\hat{\beta}_1$ points (95% CI: ___ to ___)."

Interpreting the intercept ($\hat{\beta}_0$): Often not meaningful — only interpret when $X = 0$ is realistic in context.

The LINE conditions

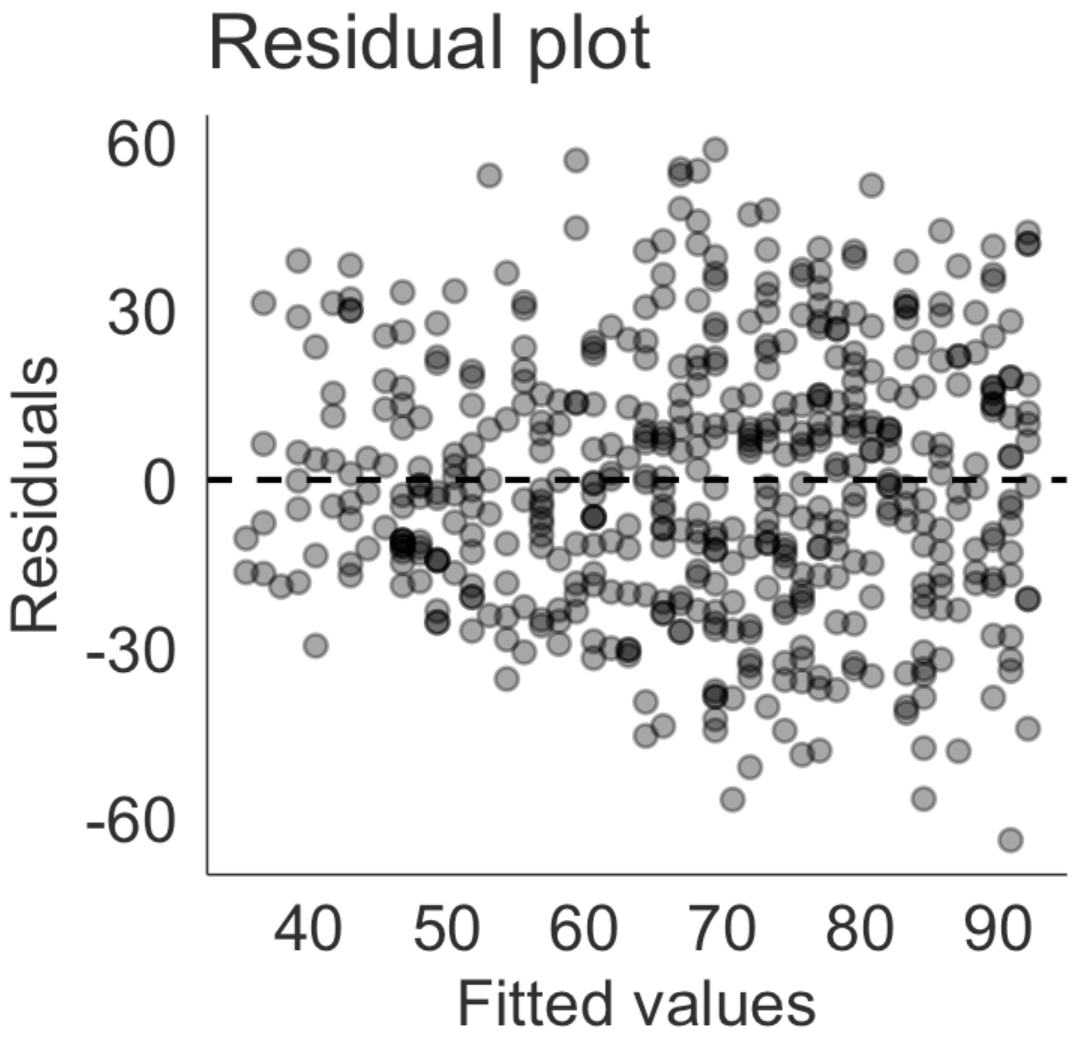
Letter	Condition	How to check
L	**L**inearity — relationship is linear	Scatterplot; residual plot
I	**I**ndependence — observations independent	Study design
N	**N**ormality — residuals approximately normal	QQ plot of residuals
E	**E**qual variance — residuals have constant spread	Residual plot

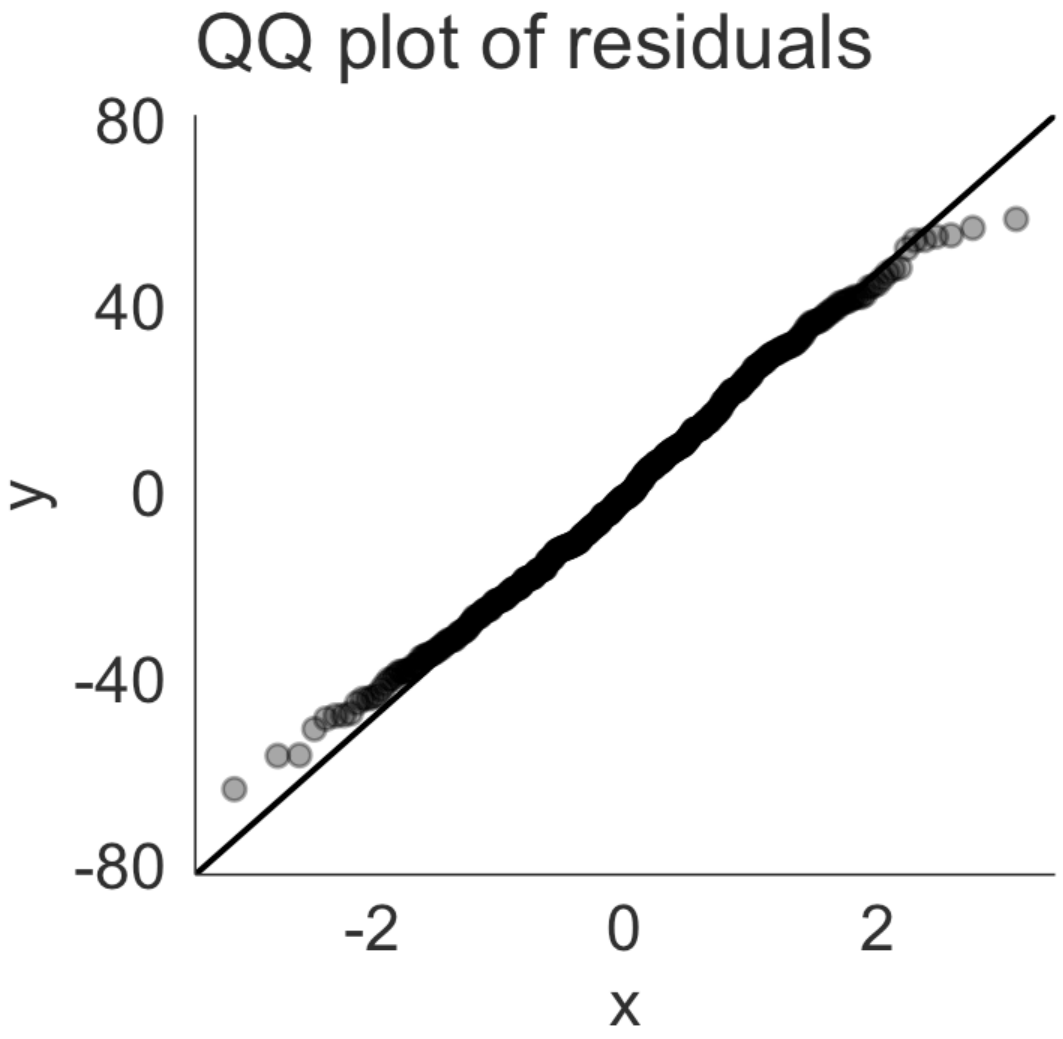
```r
aug <- augment(rfft_model)

# Residual plot (checks L and E)
ggplot(aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted values", y = "Residuals", title = "Residual plot")

# QQ plot (checks N)
ggplot(aug, aes(sample = .resid)) +
  stat_qq(alpha = 0.4) + stat_qq_line() +
  labs(title = "QQ plot of residuals")
```

Diagnostic plots for rfft_model





Reading the diagnostic plots

Residual plot (L and E)

Good: Points scatter randomly around 0; flat smoother; roughly equal spread

Red flags:

- Curve → linearity violated (L)
- Funnel → equal variance violated (E)

QQ plot (N)

Good: Points track the diagonal reference line

Red flags:

- S-shape → skewed residuals
- Heavy tails → points deviate at both ends

Note: With large n, the CLT makes moderate departures from normality less of a concern. L and E are more critical for inference.

R^2 and sigma

R^2 interpretation

"Age explains ___% of the variability in RFFT score in this sample."

R^2 ranges from 0 (model explains nothing) to 1 (perfect fit).

For SLR: $R^2 = r^2$ (square of Pearson correlation)

sigma

`sigma` = residual standard deviation = typical distance of an individual observation from the fitted line, in **units of** Y.

This captures individual-level scatter. It is much larger than `std.error` for the slope, and it is why prediction intervals are so much wider than CIs for mean response.

```r
# Model-level summary: R², sigma, n
glance(rfft_model) %>% select(r.squared, sigma, nobs)
```

```
# A tibble: 1 × 3
  r.squared sigma  nobs
      <dbl> <dbl> <int>
1     0.285  23.2   500
```

Tip

`std.error` (from `tidy()`) tells you how well you estimated the slope. `sigma` (from `glance()`) tells you how much individuals vary around the line. These measure completely different things.

Inference for the slope: six steps

Step 1: $H_0 : \beta_1 = 0$ (age tells us nothing about RFFT) vs. $H_A : \beta_1 \neq 0$

Step 2: $\alpha = 0.05$

Step 3: Check LINE conditions ✓

Step 4: $t = \hat{\beta}_1 / SE_{\hat{\beta}_1}$ — read from `tidy()` output, `statistic` column

Step 5: p-value from `tidy()` output, `p.value` column; $df = n - 2$

Step 6: Conclusion in context:

> "There is strong evidence of a negative linear association between age and RFFT score (t = ___, df = ___, p = ___). For every 1-year increase in age, RFFT score is expected to decrease by ___ points (95% CI: ___ to ___)."

CI for mean response vs. prediction interval

	Question	Accounts for
CI for mean response	Average RFFT for **all people** aged 60?	Uncertainty about where the line is
Prediction interval	RFFT for **this specific new patient** aged 60?	Uncertainty about the line **+** individual variation ($\hat{\sigma}$)

```r
new_data <- data.frame(Age = c(45, 60, 75))

# CI for mean response
predict(rfft_model, newdata = new_data, interval = "confidence")

# Prediction interval for a new individual
predict(rfft_model, newdata = new_data, interval = "prediction")
```

Important

Prediction intervals are **always wider** than CIs for mean response. More data narrows the CI (better-estimated line) but cannot eliminate individual variation — that scatter is a property of the population, not of your sample size.

Part 8: Power, Sample Size, and Synthesis

Power analysis with `pwr.t.test()`

Cohen's d: effect size for comparing two means

$$d = \frac{\mu_1 - \mu_2}{\sigma_{\text{pooled}}}$$

Example: detect a 5 mmHg difference in SBP, pooled SD = 10 → $d = 0.5$

Solve for n:

```r
pwr.t.test(n           = NULL,
           d           = 5 / 10,
           sig.level   = 0.05,
           power       = 0.80,
           type        = "two.sample",
           alternative = "two.sided")
```

Solve for power with n = 50 per group:

```r
pwr.t.test(n           = 50,
           d           = 5 / 10,
           sig.level   = 0.05,
           power       = NULL,
           type        = "two.sample",
           alternative = "two.sided")
```

Tip

Always **round up** the required n — rounding down means you're slightly underpowered.

Sample size and what it affects

When you increase n:

	Effect
ANOVA F-statistic	Increases (more power to detect group differences)
Post-hoc p-values	Decrease (more power per pairwise comparison)
Width of CIs	Narrower (smaller SE → tighter interval)

Peeking at data before choosing the test

- Running ANOVA, seeing which group performed best, then running a t-test comparing that group vs. the others is **data-driven test selection** — it inflates the Type I error rate because you used the data twice.
- The right approach: specify your comparisons **before** looking at the data.

Choosing the right test: the synthesis table

Research question	Groups	Outcome	Test	Key R function
Did the outcome change? (same subject, 2 timepoints)	1 (differences)	Continuous	Paired t-test	`t.test(d, mu = 0)`
Does mean outcome differ across 3+ groups?	3+	Continuous, normal	One-way ANOVA	`aov()`
Does a skewed outcome differ across 3+ groups?	3+	Continuous, skewed	Kruskal-Wallis	`kruskal.test()`
Is a categorical variable associated with a binary outcome?	3+	Binary	Chi-squared	`chisq.test()`
Does a binary outcome rate differ between 2 groups?	2	Binary	Two-proportion z-test	`prop.test()`
Is there a linear association between two continuous variables?	—	2 continuous	SLR / slope test	`lm()` + `tidy()`

CI and hypothesis test: two sides of the same coin

A recurring exam theme

For a two-sided test at $\alpha = 0.05$:

- If the **95% CI excludes** the null value → p-value < 0.05 → reject H_0
- If the **95% CI includes** the null value → p-value ≥ 0.05 → fail to reject H_0

What the CI adds beyond the p-value:

- The **direction and magnitude** of the effect (not just "significant or not")
- A range of **plausible values** for the true population parameter
- Clinical interpretability — a statistically significant effect may be too small to matter in practice

Good examples: any test where you computed both a CI and a p-value. Ask: what does the CI tell us about the *size* of the effect that the p-value alone cannot?

Reminders and Good Luck

Common mistakes to avoid

1. **Ignoring paired structure** — running an independent t-test on before/after data
2. **Forgetting to check assumptions** — especially ANOVA's SD ratio and chi-squared expected counts ≥ 5
3. **Misframing nonparametric hypotheses** — say "distributions are identical," not "medians are equal"
4. **Confusing σ, s, SE, and `sigma` from `glance()`** — know what each measures
5. **Using `lower.tail = TRUE/FALSE` incorrectly** — sketch the picture first
6. **Not rendering before submitting** — code that works in a chunk but breaks the render is ungraded

You are ready

- ✅ You have worked through 8 homework assignments with increasingly complex analyses
- ✅ You have practiced the six-step framework across many test types
- ✅ You have hands-on experience with every R function on this exam
- ✅ Part 7 has hints — you are not going in cold on SLR

The exam opens at 3:00 PM today. You have until Friday at 5:00 PM.

Read carefully, interpret in context, save frequently, and trust your preparation.

Good luck!

Questions?

Your Turn

What would you like to go over?

- Concepts that feel unclear?
- R functions you want to see again?
- Specific parts of the exam structure?

This is your time — ask anything.